



The Processing of Chinese Reflexives as Plain Anaphors and Intensifiers

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Abstract

This study investigates the processing of two Chinese reflexives, *ziji* ('self') and *ta-ziji* ('s/he-self'), in structural contexts that support plain anaphor and intensifier interpretations, respectively. Two self-paced reading experiments were conducted to examine the interpretive patterns. Converging evidence from post-reading acceptability judgments and reading times showed that, in contexts supporting plain anaphor readings, both reflexives exhibited locality biases. However, in structural contexts that enforce intensifier readings, the processing patterns changed. These findings highlight the different processing profiles of the two reflexives and provide new insights into how discourse and syntactic constraints interact during online reflexive resolution.

Keywords Chinese · Reflexive resolution · Plain anaphor · Intensifier

Introduction

Pronominal elements derive their meaning from the referents or *antecedents* they are linked to. Whether a referent can serve as the antecedent depends on semantic, syntactic, and discourse-pragmatic constraints that guide pronoun resolution. This study focuses on reflexive pronouns (e.g., *himself/herself*), which have played a key role in advancing our understanding of how linguistic constraints interact during real-time processing. In this study, we examine the processing profiles of two Chinese reflexives, the morphologically simple reflexive *ziji* ('self') and the morphologically complex reflexive *ta-ziji* ('s/he-self').

The processing of reflexives has produced considerable debate in the field of psycholinguistics. A key question is whether the real-time or online interpretation of reflexives strictly adheres to Principle A of the Binding Theory (Chomsky, 1981). Principle A is a locality constraint which posits that a reflexive must be linked to an antecedent within the *local domain*. Following the standard definition, we define a local domain as a finite clause containing a

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reflexive along with a clausemate subject (e.g., Chomsky, 1986; Huang et al., 2001) and refer to reflexives governed by the locality constraint as *plain anaphors* (e.g., Charnavel, 2021). For example, in (1), the local referent *Sam* in the subordinate clause is strongly favored as the antecedent of the reflexive *himself* in English. In Chinese, however, where reflexives can appear in subject position, the local domain can extend to the superordinate clause when the subordinate clause lacks an accessible clausemate subject, as shown in (2).¹

(1) John₁ said that [Sam₂ owed himself_{*1/2} another chance]_{local-domain}.

(2) [Zhāngsān gàosu wǒ zìjǐ pīpíng-le Lísì]_{local-domain}.

Zhangsan tell me self scold-ASP Lisi.

Zhangsan told me he scolded Lisi.

While substantial evidence suggests that the locality constraint plays a crucial role in real-time reflexive resolution in English (e.g., Sturt, 2003; Dillon et al., 2013; Felser et al., 2009; Cummings & Felser, 2013; Cummings & Sturt, 2014; Fujita & Yoshida, 2024), the picture is more complex for the processing of Chinese reflexives. One complication is that the two reflexive forms, *ziji* and *ta-ziji*, appear to be differentially constrained by the locality constraint during online resolution. Some studies report that *ziji* exhibits a stronger locality bias than *ta-ziji* (e.g., Dillon et al., 2016; Wang, 2017), whereas more recent findings suggest the opposite pattern, with *ta-ziji* showing stronger locality effects (e.g., Lyu & Kaiser, 2023; Ma & Wu, ms.). Adding to this complexity, Chinese reflexives are particularly sensitive to discourse-level constraints. For example, when the non-local referent is non-topical, local antecedents are favored (e.g., Li & Zhou, 2010; Dillon et al., 2014, 2016; Jäger et al., 2015; Chang et al., 2020; Lyu & Kaiser, 2023), suggesting that, by default, *ziji* and *ta-ziji* are interpreted as plain anaphors; but when a non-local referent is topically prominent, it can ‘attract’ coreference (e.g., Lyu & Kaiser, 2021, 2023), thus overriding the locality constraint.

Building on previous findings, the present study pursues two goals. First, we test whether *ziji* and *ta-ziji* differ in their sensitivity to the locality constraint in structural contexts where the non-local referent is not discourse-prominent and thus favor plain anaphor readings. We focus on reflexives in subject position as the debate primarily arises in that domain. Second, we investigate how and when discourse factors influence the online processing of Chinese reflexives by introducing contrastive focus into sentence structure. Specifically, we used the *shi*-cleft focus construction to elicit a contrastive or intensifier reading of the reflexive. To these ends, we conducted two self-paced reading experiments. Previewing our results, we

¹We note that, as discussed in prior work (i.e., Huang et al., 2001, 2009), the extension of the binding domain to a higher clause is compatible with the standard definition of the binding domain, which requires the presence of an accessible SUBJECT for the reflexive. In the case of subject reflexives, such a SUBJECT is not available within the embedded clause and can only be found in a higher clause. Crucially, Huang et al. (2001: 327) argued that in (2), if *ziji* were in a non-local relationship with Zhangsan, it would function as a logophoric (i.e., perspective-sensitive) reflexive. However, the logophoric analysis of *ziji* is untenable as the first-person pronoun would block non-local binding due to the Speech Empathy Hierarchy (e.g., Kuno, 1987) prioritizing the speaker’s own perspective than those of others. The absence of such a blocking effect therefore indicates that *ziji* here functions as a plain anaphor. In other words, when *ziji* in subject position lacks a clausemate antecedent, the local domain extends to the higher finite clause. We adopt Huang and Liu’s definition of locality for subject reflexives in this paper. A fuller discussion is provided in the supplementary materials on the OSF repository.

find suggestive evidence that *ta-ziji* shows an earlier locality bias than *ziji* when parsed as plain anaphors, and that contrastive focus modulates reflexive processing at early stages. Taken together, these findings suggest that online reflexive resolution in Chinese is jointly guided by syntactic and discourse-pragmatic cues.

Research Background

The Processing of Plain Anaphors

Generalizing over a large body of research, three uses of Chinese reflexives can be identified (e.g., Pan, 1998, 2001; Huang et al., 2001; Sperlich, 2020). The first distinction is a functional one and concerns whether the reflexive is used contrastively or non-contrastively.² The second distinction applies in the non-contrastive case, where the reflexive may function either as a plain anaphor subject to the locality constraint or as a logophor exempt from it (e.g., Yu, 1992; Huang, 2000; Huang et al., 2001).

The term *logophor* originally referred to a specific type of pronouns in certain West African languages that obligatorily refer to the agent of speech and thought (Hagège, 1974; Clements, 1975). But it has since been extended to cover long-distance reflexives (e.g., Sells, 1987), which can likewise refer to perspectival centers in non-local domains (e.g., Pollard & Sag, 1992; Reinhart & Reuland, 1993; Huang et al., 2001). In fact, Charnavel et al. (2017: 54) proposed that “the antecedent for a long-distance reflexive must be an internal pivot” (i.e., perspectival center). In this paper, we adopt this widely held perspective-based view: non-local antecedents can be accessed via perspectival mechanisms, although additional, mutually non-exclusive factors such as verb bias may also play a role (see discussion in e.g., Pollard & Xue, 1998).

These three uses of reflexives may differ in the distribution of their antecedents. A contrastive reflexive or intensifier can take either a local or non-local antecedent, depending on which antecedent is contrasted with alternative referents (see Sect. 2.2 below). Similarly, a logophoric reflexive may also refer to a local or non-local antecedent, provided that its antecedent is a perspectival center. By contrast, a plain anaphor is locally bound. Therefore, reflexives in Chinese are in principle ambiguous among these uses. However, the contrastive and logophoric uses are more marked, conditioned by prosodic and/or contextual cues, and thus less frequent. For example, a reflexive intensifier presupposes the evocation of an alternative referent for comparison and bears stress (e.g., Pan, 1998; Hole, 2008; Wang, 2011), while a logophoric reflexive depends on the presence of a discourse-prominent referent whose perspective the speaker adopts. For this reason, in sentences such as (3) (Lu, 2011: 23), researchers generally assume that *ziji* or *ta-ziji* is preferably interpreted as a plain anaphor (e.g., Lu, 2011; Wang, 2017; Ma & Wu, 2025) because there is neither an alternative referent to be contrasted with nor a prominent perspectival center in the discourse.

² In situations where the contrastive reflexive expresses the perspective of an attitude holder, the contrastive use can be considered as one specific type of logophoric use. But since the contrastive use does not necessarily express the perspective of the referent (e.g., ‘Lisi will finish the job all by himself.’), at least in Chinese, we keep the contrastive and logophoric uses separate.

- (3) Yóuchāi₁ xuānchéng dīnghù₂ zǒng huáiyí zìjǐ/tā-zìjǐ méiyǒu ànshí tóudi/shōudào yóujiàn mailman claim subscriber always suspect self/he-self not on time deliver/receive mail.

The mailman₁ claimed that the subscriber₂ always suspected that he did not deliver₁/receive₂ the mail on time.

In (3), as argued by Huang et al., (2001), the reflexive in subject position stands in a local relationship with the subject of the higher clause ('the subscriber'). However, the reflexive can flexibly refer to either the structurally local 'subscriber' or the non-local 'mailman,' depending on the meaning of the verbs in the minimal finite clause. When the verb meaning ('deliver') is congruent with the non-local 'mailman' as the antecedent, non-local binding is preferred; when the verb meaning ('receive') is congruent with the local 'subscriber' as the antecedent, local binding is preferred. Assuming that both reflexives show local biases during real-time processing due to a preferred plain anaphor interpretation, we might expect longer reading times when the verb meaning is incongruent with the local referent. However, prior research has not reached consensus on the locality biases associated with the processing of *ziji* and *ta-ziji*.

In two self-paced reading experiments, Lu (2011) found that in sentences like (3), *ziji* shows a non-local bias while *ta-ziji* shows a local bias: at the disambiguating verb ('deliver/receive'), locally congruent verbs elicited longer reading times than non-locally congruent verbs in the case of *ziji* while the opposite patterns emerged for *ta-ziji*. However, as Ma and Wu (2025) suggest, Lu's findings may allow for alternative interpretations. The predicate (e.g., 'suspect') preceding the subject reflexive semantically biases the matrix subject as the antecedent, since one typically suspects others' actions rather than one's own. It is therefore possible that participants relied on the predicate information to anticipate coreference with the non-local subject before encountering the reflexive itself. Crucially, in our view, this experimental design did not reveal whether native Chinese speakers preferred local or non-local binding at the point when the reflexive was first encountered. The binding bias effect observed at the disambiguating verb, several words after the reflexive, only reflects interpretation preferences at a later processing stage.

The question of whether *ziji* and *ta-ziji* exhibit similar locality biases was similarly pursued by Wang (2017) and, more recently, by Ma and Wu (2025). Both studies employed self-paced reading to investigate the processing of reflexives in subject position in similar sentence structures, as shown in (4a–b). Following Lu (2011), they used the verb congruency diagnostic – where longer reading times elicited by non-locally congruent verbs indicate a locality bias – to probe participants' binding preferences at the disambiguating verb. Wang (2017) reported a locality bias for *ziji* but no bias for *ta-ziji*. By contrast, Ma and Wu (2025), using a considerably larger sample, found comparable locality biases for both reflexives. However, both studies placed the reflexive inside a relative clause and after a transitive verb (e.g., 'prefer'), which might lead participants to initially parse the reflexive as an object.

- (4) a. Local condition.

Tīngzhòng₁ biǎoshì yǎnjiǎngzhě₂ piānài zìjǐ/tā-zìjǐ qiántiān yǎnjiǎng₂ de zhùtí.

audience state orator prefer self/he-self yesterday speak COMP topic.

The audience₁ said that the orator₂ preferred the topic he talked₂ about yesterday.

b. Non-local condition.

Yǎnjiǎngzhě₁ biǎoshì tīngzhòng₂ piānài zìjǐ/tā-zìjǐ qiántiān yǎnjiǎng₂ de zhǔtí.

orator state audience prefer self/he-self yesterday speak COMP topic.

The orator₁ said that the audience₂ preferred the topic he talked₁ about yesterday.

We note that the three studies discussed above all investigated the processing of Chinese reflexives in subject position, a question we similarly address in this study to clarify the debate. In contrast, studies examining reflexives in direct object position (e.g., Gao et al., 2005; Dillon et al., 2014, 2016; Chang et al., 2020) or as possessors modifying direct objects (e.g., Chen et al., 2012; Lyu & Kaiser, 2023) have consistently reported a locality bias for both reflexives. Among these, Dillon et al.'s (2016) self-paced reading study is especially relevant, as it directly compared the processing of *ziji* and *ta-ziji*. Using an animacy-mismatch paradigm, they manipulated binding distance by making either the local or non-local antecedent animate while keeping the other referent inanimate, as illustrated in (5a–b). Reading slowdowns triggered by animacy mismatches between the reflexive and a potential antecedent were taken as evidence for binding attempts. Their results showed that although both reflexives exhibited a locality bias at the reflexive region – non-local binding elicited reading slowdowns – the spillover effect at the next word was stronger for *ziji* than for *ta-ziji*. This finding may point to subtle differences in the processing of the two reflexive forms.

(5) a. Local condition.

Méití bàodào de nà-ge nǚcáifeng shàng-ge xīngqī bǎ zìjǐ/tā-zìjǐ nòngshāng-le.

media report COMP that-CL seamstress last-CL week BA self/he-self hurt-ASP.

The seamstress that the media reported last week hurt herself last week.

b. Non-local condition.

Zhāng tàitai guānggù de nà-jiā shízhuāngdiàn shàng-cì bǎ zìjǐ/tā-zìjǐ nòngshāng-le.

Mrs. Zhang visit COMP that-CL boutique last-time BA self/she-self hurt-ASP.

The boutique that Mrs. Zhang visited hurt herself last time.

To summarize, prior psycholinguistic research on the processing of *ziji* and *ta-ziji* in direct object and object-modifying possessor positions has yielded consistent results regarding the locality bias. However, findings on reflexives in subject position remain inconclusive

and that the experimental design (i.e., verb congruency diagnostic) could not immediately detect the binding preferences at the onset of the reflexive. Furthermore, direct comparisons between the processing profiles of the two reflexive forms are still limited. We seek to address these open questions in this study.

The Processing of Intensifiers

The previous subsection reviewed prior psycholinguistic research on the processing of Chinese reflexives as plain anaphors. In this section, we turn to reflexives used as intensifiers. We first introduce intensifier reflexives in Chinese and review previous experimental work on intensifiers. We then situate the study of intensifiers within processing theories that address the relative accessibility of syntactic versus discourse-pragmatic constraints.

In the theoretical literature, the function of a reflexive intensifier is to evoke alternative referents which the reflexive's antecedent is contrasted with (e.g., König & Siemund, 2000). For example, Chinese *ziji* as an intensifier often appear in three positions: as adnominal intensifiers, adverbial intensifiers, and attributive intensifiers (e.g., Hole, 2008; Wang, 2011), as illustrated in (6a–c). In (6a), the sentence emphasizes that Zhangsan himself cannot come while evoking an alternative, his father, who can. In (6b), Zhangsan's intention to solve the problem by himself is contrasted with alternative scenarios of having others solve it for him. In (6c), Zhangsan's action of taking his own umbrella is contrasted with the alternative situation of him taking someone else's umbrella. Across these cases, the reflexive intensifier puts the antecedent of the reflexive in focus while evoking (implied) alternative referents for contrast.

- (6) a. Zhāngsān zìjǐ bù néng lái, dàn tā fùqīn néng lái. (adnominal intensifier)

Zhangsan self not can come but his father can come

Zhangsan himself cannot come, but his father can.

- b. Zhāngsān xiǎng zìjǐ jiějué zhè-ge wèntí. (adverbial intensifier)

Zhangsan want self solve this-CL problem.

Zhangsan wanted to solve the problem by himself.

- c. Zhāng Sān zhǐ dàizǒu-le zìjǐ de sǎn, méi ná qí tā rén de sǎn. (attributive intensifier)

Zhangsan only take-ASP self GEN umbrella not take other person GEN umbrella.

Zhangsan only took his own umbrella and did not take other peoples' umbrellas.

However, intensifier reflexives can also occur in subject position, especially when preceded by a focus particle in a cleft structure, as in (7). In such cases, *shi* functions as a focus particle, marking the adjacent constituent as the sentence focus (e.g., Cheng, 2008; Paul & Whitman, 2008). The reflexive following *shi* is therefore obligatorily focused and necessarily

receives a contrastive reading with contrastive stress. Notably, as Pan (1998: 788) observed, *ta-ziji* used as an intensifier can be bound not only across clause boundaries but even across sentences. In (7), both the local and non-local subject can serve as antecedents, since either referent can in principle be contrasted with an alternative set of individuals.

(7) Zhāngsān biǎoshì Lǐsì rěnwéi shì zìjǐ/tā-zìjǐ bǎ shìqíng gǎozá-le.

Zhangsan state Lisi think FOC self/he-self BA matter mess.up-ASP.

Zhangsan stated that Lisi thought it was him (and not anybody else) who messed things up.

Crucially, Baker (1995) proposed that the interpretation of reflexive intensifiers is constrained by two linguistic conditions. The *contrastiveness condition* requires that intensifiers appear only in contexts where emphasis or contrast is intended, while the *condition on relative prominence* stipulates that the antecedent of an intensifier must be the referent that is more prominent (see similar views in Pan, 1998; Liu, 2008). Together, these conditions predict that in the *shi*-cleft construction, reflexives seek prominent antecedents. Because prominence can be achieved in multiple ways, including topicality and syntactic locality (e.g., Pan, 1998; Hu & Pan, 2002), both the local and non-local subjects in sentences like (7) can serve as potential antecedents, in contrast to sentences like (3) where local binding is often preferred. For this reason, we may anticipate that when interpreted as intensifiers, reflexives may show attenuated locality biases.

While previous experimental work has paid relatively little attention to intensifiers or their differences compared to plain anaphors, Kaiser and Runner (2008) compared the interpretation of plain anaphors and intensifiers in German and Dutch. In these languages, the two reflexive forms are distinct: in German, *sich* is the plain anaphor and *sich selbst* is the intensifier; in Dutch, *zichzelf* is the plain anaphor and *hemzelf* the intensifier. Using a forced-choice judgment task, Kaiser and Runner (2008) investigated native speakers' interpretative preferences for these forms. They found that plain anaphors and intensifiers preferred local antecedents to similar degrees in German, but in Dutch, the intensifier showed a predicted weaker locality bias than plain anaphors. This suggests that contrastive focus modulates the interpretation of reflexives in Dutch, although its influence in German is less clear. However, because Kaiser and Runner's (2008) study was offline, it did not reveal how the locality constraint interacts with discourse focus during real-time processing, an issue that is central to debates on reflexive resolution and the focus of the present study.

Turning to real-time processing, the question of whether intensifiers adhere to the locality constraint has rarely been examined. While it is conceivable that contrastive focus might modulate the locality bias during real-time processing given Baker's (1995) condition on relative prominence, a rich body of research exploring the interaction between syntactic locality and non-syntactic cues – including gender, number, and discourse topicality – suggests that syntactic locality may play a predominant role. This line of work, focusing primarily on English reflexives, indicates that the locality constraint takes precedence over non-syntactic cues (e.g., Sturt, 2003; Felser et al., 2009; Cummings & Sturt, 2013; Dillon et al., 2013; Jäger et al., 2020; Fujita & Yoshida, 2024; but see Kaiser et al., 2009; Patil et al., 2016), a position we refer to as the *syntax-first account*.

For Chinese reflexives, however, the relative accessibility of syntactic and non-syntactic cues remains more contested (e.g., Jäger et al., 2015; Chang et al., 2020), particularly regarding discourse-level cues (e.g., Lyu & Kaiser, 2023, 2024; Zhao & Sperlich, 2025). Discourse properties such as topic prominence and perspective-taking have been shown to play important roles during online reflexive resolution in Chinese, supporting a *multiple-constraints account* in which diverse linguistic cues are immediately available to the parser. For instance, Lyu and Kaiser (2023) found in a self-paced reading study that when the non-local subject was not a discourse topic, reflexive resolution adhered to the locality constraint; but when the non-local subject was topically prominent, readers showed immediate sensitivity to this discourse cue. Similarly, the presence of the contrastive focus particle *shi* may lead the parser to extend the search domain and prioritize prominent antecedents in discourse. While we think that the relative weight of syntactic and non-syntactic cues may vary cross-linguistically (e.g., Mertzen et al., 2023) and may depend on the specific cues involved, examining how discourse focus interacts with the locality constraint in Chinese can offer new insights into the mechanisms underlying sentence comprehension.

More recently, in a self-paced reading study, Zhao and Sperlich (2025) made some initial progress in examining the role of contrastive focus in online reflexive resolution. They constructed contrastive contexts in which a reflexive could plausibly receive an intensifier interpretation, as in (8). Assuming that reflexives in such contexts were interpreted as intensifiers – thus referring to the contrasted, non-local antecedent – they found that *ziji* and *ta-ziji* elicited similar reading times, which was interpreted as suggesting that the two reflexives were similarly influenced by contrastive focus. Their study thus represents an important step towards understanding the role of contrastive focus in real-time reflexive resolution.

- (8) a. ‘Huangyi₁’s younger brother and Zhangyu₂’s younger brother are classmates.
Zhangyu₂’s younger brother has poor health and often takes medicine.’ (context sentence).

- b. Zhāngyǔ₁ shuō Huángyì₂ tóngqíng zìjǐ₁-de/tā-zìjǐ₁-de dìdì-le. (critical sentence)

Zhangyu say Huangyi sympathize self-GEN/he-self-GEN brother-ASP.

Zhangyu₁ said that Huangyi₂ sympathized with his₁ younger brother.

However, two open questions remain. First, our native speaker judgments suggest that Zhao and Sperlich’s contrastive contexts may not necessarily elicit strong expectations for contrastive readings: readers could also adopt a non-contrastive interpretation, particularly when the contrasted referent (e.g., ‘younger brother’) appears after the reflexive in a self-paced reading task. It is thus unclear how readers could predict the identity of the upcoming contrasted referent (‘younger brother’) before its onset. Moreover, the reflexives in (8) tend to be initially pronounced without prosodic stress according to the informants we consulted, raising the possibility that readers did not always construe the reflexives as intensifiers during the initial parse. Second, the study did not include a diagnostic measure for assessing binding preferences, making it difficult to determine whether and when contrastive focus affects processing. Given these considerations, Zhao and Sperlich’s (2025: 8) conclusion

that “*ziji* with its long-distance binding potential, is effortlessly guided to the long-distance antecedent, and *ta-ziji* responds to the context that works with its emphatic interpretation” needs to be tested empirically.

To address these open issues, we tested sentences with the *shi*-cleft structure, as illustrated in (7), to probe the accessibility of contrastive focus during real-time processing. Crucially, as noted above, the constituent following *shi* obligatorily receives contrastive focus, ensuring that the reflexive is parsed as an intensifier that can felicitously probe for an antecedent outside the local domain (e.g., Baker, 1995; Pan, 1998). Additionally, we compare the processing of Chinese reflexives as plain anaphors and intensifiers, thereby examining whether reflexive resolution shows form-specific differences (e.g., Dillon et al., 2016; Wang, 2017) in these uses.

Aims of the present study

We report two self-paced reading experiments, paired with end-of-sentence acceptability judgments, to examine the processing of *ziji* and *ta-ziji* in structural contexts biasing plain anaphor readings and in *shi*-cleft constructions enforcing intensifier readings. We have two overarching goals.

First, we aim to explore the processing similarities and differences between *ziji* and *ta-ziji* as plain anaphors in subject position. Previous research has yielded inconsistent findings regarding this position (e.g., Lu, 2011; Wang, 2017; Ma & Wu, 2025, ms.). Unlike earlier studies, which could only detect locality biases at the post-reflexive verb region, we followed Dillon et al. (2016) in adopting an animacy-mismatch paradigm, allowing locality effects to be detected immediately at the onset of the reflexive.

Second, we aim to shed light on the relative accessibility of contrastive focus and syntactic locality in online processing by examining intensifiers in *shi*-cleft constructions. Although non-local binding is possible, early-stage processing may nevertheless be dominated by syntactic constraints (e.g., Cunnings & Sturt, 2014), given considerations of computational economy, i.e., non-local referential dependencies via the discourse route are more costly to establish (e.g., Reuland, 2011). By doing so, we view our study as a step toward clarifying the role of discourse cues in reflexive resolution, building on accumulating evidence that discourse factors play prominent roles in real-time reflexive resolution in Chinese.

Experiment 1: Processing of Plain Anaphors

Participants

One hundred and four native Chinese speakers from the Peking University community participated in the experiment. All participants had normal or corrected-to-normal vision. Informed consent was obtained before the experiment. Participants received compensation for participation.

To estimate statistical power, we conducted a simulation-based power analysis using the *mixedpower* function (Kumle et al., 2021) in R, based on the mixed-effects models for reading times (see Sect. 4.4). The analysis indicated that the power to detect a fixed

effect ranges between 0.46 and 0.71, assuming an effect size between 40–70ms. Given the observed locality effect at the critical region in Experiment 1, the power is plausibly close to 0.7. This estimate is broadly consistent with the benchmark offered by Brysbaert (2019: 25), who notes that for a medium-to-small effect size (Cohen's $d=0.4$) in a 2×2 factorial design, a sample of 110 participants is required to achieve a power of 0.8.

Design and Materials

The factors Distance (local/non-local) and Reflexive (*ziji/ta-ziji*) were crossed in a 2×2 factorial design. Following Dillon et al. (2016), we used animacy congruency to bias the interpretation toward local or non-local dependencies, based on the assumption that animate referents are more felicitous antecedents than inanimate ones. This animacy manipulation is thus an essential feature of the experimental design. To verify its effectiveness, we conducted a separate forced-choice norming study with 20 native speakers using the target items. The results confirmed that the manipulation reliably elicited the intended local and non-local binding preferences, with the proportions of animate referent choices across conditions ranging from 82% to 100% (see supplementary materials).

The factor Distance refers to the referential dependency length between the reflexive and its antecedent, and Reflexive refers to the type of reflexives under investigation. An example set of target items is shown in (9), with the numbered subscripts indicating presentation regions in the self-paced reading task. Note that for the most embedded clause, we followed Dillon et al. (2016) in using the BA-construction to create more spillover regions so as to better capture the spillover effects.³ The particle BA has the grammatical function of preposing the object before the verb (see Huang et al., 2009: 154–196 for an overview).

(9) (a) Local binding.

Gōngsī₁ biǎoshì₂ Xiǎolì₃ rènwéi₄ zìjǐ/tā-zìjǐ₅ bǎ₆ yánjiū₇ jìhuà₈ dǎluàn-le.₉

company state Xiaoli think self/she-self BA research plan disrupt-ASP.

The company stated that Xiaoli believed she had disrupted the research plan.

(b) Non-local binding.

Xiǎolì₃ biǎoshì₂ gōngsī₁ rènwéi₄ zìjǐ/tā-zìjǐ₅ bǎ₆ yánjiū₇ jìhuà₈ dǎluàn-le.₉

Xiaoli state company think self/she-self BA research plan disrupt-ASP.

Xiaoli stated that the company believed she had disrupted the research plan.

To balance the pronoun gender in *ta-ziji*, half of the referents were male and half female. Twenty-four sets of target items were distributed into four lists using a Latin square design.

³ We acknowledge an anonymous reviewer's comment that the BA-construction might incur higher processing costs, potentially complicating the reflexive resolution process. We regard this as an important venue for future research.

As a result, each participant read six items in each condition. The target items were mixed with 36 fillers and presented to participants in random order.⁴ Because participants were asked to provide end-of-sentence acceptability judgments, we included both grammatical and ungrammatical fillers to ensure participants paid attention to the experiment.

Procedure

The experiment was run on PCIBex (Zehr & Schwarz, 2018). Participants pressed the spacebar to read sentences region by region in a moving window paradigm. Each key press revealed a new region while masking the previous region. Upon finishing the sentence, participants rated the acceptability of the sentence on a 1–7 Likert scale, with higher numbers indicating higher acceptability. The left and right ends of the scale were labeled ‘completely unacceptable’ and ‘completely acceptable,’ respectively.

We used an acceptability judgment task rather than a forced choice task because explicitly probing the antecedent of the reflexive could reveal the purpose of the experiment, potentially disrupting naturalistic reading and encouraging metalinguistic strategies. Furthermore, given the animacy design, a forced choice task may not be particularly informative about participants’ preferences for local versus non-local binding, as responses could be driven primarily by the animacy of the antecedents rather than binding preferences.

Data Analysis

We analyzed post-reading acceptability judgments using mixed-effects ordinal regression (*clmm* package, Christensen, 2023) and reading times (RTs) using mixed-effects linear regression (*lmer* package, Bates et al., 2015) in R (R Core Team, 2024). Prior to analysis, we excluded 4 participants who did not pay close attention to the experiment: one participant whose RTs were below 100ms and three participants whose mean acceptability of grammatical fillers was lower than that for ungrammatical fillers, which left 100 participants for final analysis. The fixed effects Distance (Local: +0.5, Non-local: −0.5) and Reflexive (*Ziji*: +0.5, *Ta-ziji*: −0.5) were sum-coded. Random effects included intercepts and slopes grouped by participants and items. Statistical models were fitted using maximal random-effects structures as recommended by Barr et al. (2013); simpler models were considered only if the maximal model failed to converge or produced singular fits. We adopted a *p*-value of 0.2 as the threshold for model comparisons, as the conventional *p*-value of 0.05 over-penalizes model complexity (Matuschek et al., 2017).

For the analysis of acceptability judgments, the rating scores were z-transformed prior to analysis. When analyzing reading times, we followed a common two-step practice (see e.g., Marsden et al., 2018) by excluding outliers below 100ms and above 4000ms, as well as RTs more than three standard deviations (SDs) above the mean by condition and region. This affected 3.30% of the original data. We fitted mixed-effects linear regressions based on log-transformed RTs. *T*-values were converted to *p*-values using the Satterthwaite approximation as implemented by the *lmerTest* package (Kuznetsova et al., 2015).

⁴An anonymous reviewer notes that more fillers should be included to avoid potential adaptation effects which could weaken the locality effect. However, given that we used structurally diverse fillers and observed predicted effects, we think this issue is unlikely to affect our main conclusions.

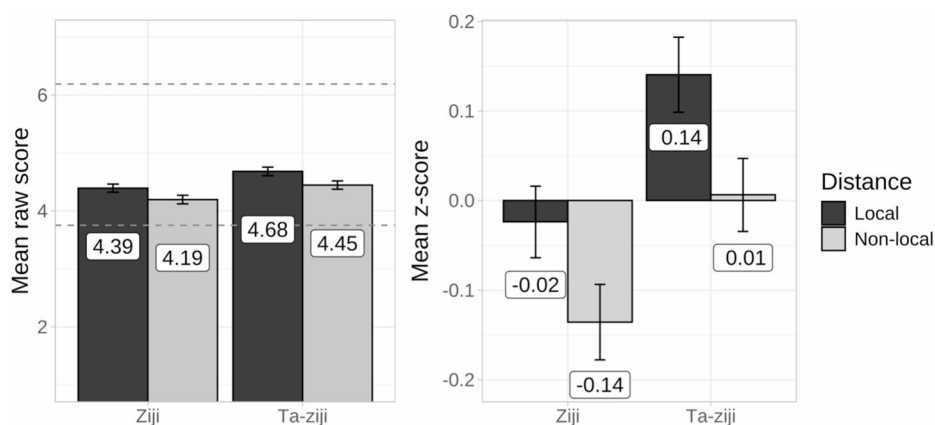


Fig. 1 Mean acceptability ratings for target conditions in Experiment 1. The left panel displays raw scores; the right panel displays z-transformed scores. Error bars indicate standard errors. The two dashed lines on the left panel represent the mean ratings for grammatical and ungrammatical fillers, respectively

Table 2 Summary of statistics for acceptability judgments in Experiment 1 (*: $p < 0.05$)

	β	SE	t	p
Distance	0.32	0.09	3.62	<0.001*
Reflexive	-0.44	0.15	-2.89	0.004*
Distance $\times$ Reflexive	-0.12	0.15	-0.81	0.42

Model structure: $\text{clmm}(\text{AJ} \sim \text{Distance} * \text{Reflexive} + (1 + \text{Reflexive} | \text{Item}) + (1 + \text{Reflexive} + \text{Distance} | \text{Participant}), \text{data})$

Results

Acceptability Judgments

We first analyze participants' post-reading acceptability judgments to assess whether *ziji* and *ta-ziji* tend to be interpreted as plain anaphors bound by local referents. Figure 1 shows participants' mean acceptability ratings for the target conditions on both raw and z-transformed scales. Table 2 summarizes the statistics.

Statistical analysis revealed a main effect of Distance ($p < 0.005$) as local binding was more acceptable than non-local binding, as well as a main effect of Reflexive ($p < 0.005$) due to a processing advantage for *ta-ziji* compared to *ziji*. The locality bias suggests that participants generally preferred the syntactically local antecedent, indicating that both reflexives were primarily interpreted as plain anaphors subject to the locality constraint. The relative advantage for *ta-ziji*, however, was unexpected. Although not crucial for our purposes, we speculate that this processing advantage may be related to reduced ambiguity: unlike *ziji*, which allows more flexible coreference in usage, *ta-ziji* exhibits more rigid binding (e.g., Huang et al., 2001), thereby limiting ambiguity in interpretation. Reduced ambiguity, in turn, may have lowered processing demands. Consistent with this possibility, as shown in the RT analysis below, participants appeared to delay reference resolution for *ziji* but resolved the reference of *ta-ziji* immediately. The Distance $\times$ Reflexive interaction was not significant.

Table 2 Summary of statistics for log-transformed RTs in Experiment 1 (*: < 0.05)

Region	Distance				Reflexive				Distance x Reflexive			
	β	SE	t	p	β	SE	t	p	β	SE	t	p
'Xiaoli/company'	0.01	0.01	0.99	0.33	-0.001	0.01	-0.08	0.94	0.04	0.02	1.76	0.08
'state'	-0.008	0.01	-0.67	0.50	0.007	0.01	0.63	0.53	-0.02	0.02	-0.80	0.43
'company/Xiaoli'	0.002	0.01	0.14	0.89	0.006	0.01	0.51	0.61	0.01	0.02	0.44	0.66
'think'	-0.01	0.01	-0.72	0.47	-0.03	0.01	-1.81	0.07	0.03	0.03	1.14	0.25
(ta-)iji	-0.04	0.02	-2.23	0.03*	-0.07	0.02	-3.45	<0.001*	0.06	0.04	1.70	0.09
BA	-0.02	0.01	-1.39	0.16	-0.04	0.01	-2.65	0.008*	0.07	0.03	2.55	0.01*
'research'	-0.01	0.01	-0.99	0.32	-0.006	0.01	-0.47	0.64	0.04	0.03	1.34	0.18
'plan'	-0.007	0.01	-0.62	0.54	0.006	0.01	0.50	0.62	0.04	0.02	1.57	0.12
'screw up'	-0.04	0.02	-1.80	0.08	0.04	0.02	1.70	0.09	-0.10	0.05	-2.12	0.04*

Model structure:

- Region 1: lmer(logRT ~ Distance * Reflexive + (1 + Distance | Item) + (1 + Reflexive | Participant), data)
- Region 2: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)
- Region 3: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 + Distance | Participant), data)
- Region 4: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)
- Region 5: lmer(logRT ~ Distance * Reflexive + (1 + Reflexive | Item) + (1 | Participant), data)
- Region 6: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)
- Region 7: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)
- Region 8: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)
- Region 9: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)

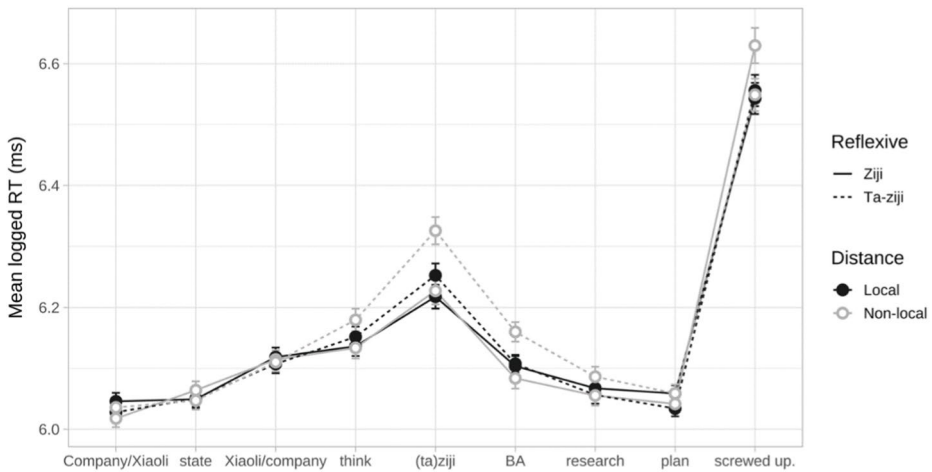


Fig. 2 Mean log-transformed RTs by region and condition in Experiment 1. Error bars represent standard errors

Reading times

Figure 2 displays the participants' mean log-transformed RTs by region and condition.⁵ See Table 2 for statistics. Before the critical reflexive region, no effects reached significance. At the reflexive, the analysis revealed main effects of Distance ($p=0.03$) and Reflexive ($p<0.001$). The main effect of Distance indicates that non-local binding elicited longer RTs overall, consistent with the view that reflexives in Chinese are by default parsed as plain anaphors when local and non-local subjects are similarly prominent at the discourse level (e.g., Chen et al., 2012; Jäger et al., 2015; Dillon et al., 2016; Lyu & Kaiser, 2023; Ma & Wu, 2025). The main effect of Reflexive partly reflects the syllabic length of *ta-ziji*, which yielded longer RTs than *ziji*. Similar reflexive-type effects have been reported in previous studies and are typically interpreted as reflecting differences between the reflexive forms themselves rather than differences in binding preferences (e.g., Dillon et al., 2016; Wang, 2017).

Interestingly, although the Distance $\times$ Reflexive interaction was only marginal and thus does not provide strong evidence for a form-specific difference, visual inspection nonetheless suggests that the two reflexives appear to differ in their locality biases. To further explore this pattern, we conducted planned pairwise comparisons using the R package *emmeans* (Lenth, 2023). Comparisons showed that the locality bias was significant for *ta-ziji* ($\beta = -0.07$, $SE=0.03$, $t = -2.82$, $p=0.004$), but not for *ziji* ($\beta = -0.01$, $SE=0.03$, $t = -0.36$, $p=0.72$). Thus, upon encountering the reflexive, participants appeared to immediately resolve the ambiguity of *ta-ziji* but did not show any bias for the processing of *ziji*.

At the first spillover region (i.e., BA), we observed a main effect of Reflexive ($p<0.01$) and a significant Distance $\times$ Reflexive interaction ($p=0.01$). The effect of Reflexive, however, was not significant. Pairwise comparisons revealed a significant locality bias for the processing of *ta-ziji* ($\beta = -0.06$, $SE=0.02$, $t = -2.84$, $p<0.005$), but not for *ziji* ($\beta=0.002$,

⁵ We follow an anonymous reviewer's suggestion in displaying log-transformed RTs (see e.g., Lakhani & Schwarz, 2024, for similar practice). The raw RT plots are available on the OSF repository.

$SE=0.02$, $t=0.80$, $p=0.42$), which strengthens the pattern observed at the critical reflexive region. At the second (e.g., ‘research’) and third (e.g., ‘plan’) spillover regions, no effect was significant. At the final region, however, we found a significant Reflexive $\times$ Distance interaction ($p<0.05$). Pairwise comparisons indicate that this effect was primarily driven by a locality bias in the processing of *ziji* ($\beta=-0.09$, $SE=0.03$, $t=-2.73$, $p<0.01$); *ta-ziji* did not exhibit a locality bias at this region ($\beta=0.007$, $SE=0.03$, $t=0.23$, $p=0.82$).⁶ Taken together, these results seem to suggest that the locality bias for *ta-ziji* emerged earlier than that for *ziji*.

Discussion

Experiment 1 investigates whether the two Chinese reflexives in subject position differ in their locality biases. Previous studies have not reached a consensus on this issue (e.g., Lu, 2011; Wang, 2017; Ma & Wu, 2025). By combining reading time data with post-reading acceptability judgments, we aim to contribute to this debate. Unlike previous studies relying on verb congruency manipulations to detect binding preferences, we employed an animacy mismatch paradigm, allowing binding preferences to be detected as soon as the reflexive is encountered.

The acceptability judgments showed that participants uniformly preferred local over non-local binding for both *ziji* and *ta-ziji*. These offline results thus support the view that both reflexives are constrained by locality (e.g., Chen et al., 2012; Jäger et al., 2015; Dillon et al., 2014, 2016; Ma & Wu, 2025). The online measure yielded converging locality effects, indicating that long-distance binding is generally more costly than local binding, in part because it may require perspective-taking at the syntax-discourse interface. This pattern is consistent with Reuland’s (2011) proposal that reflexive resolution within the syntax module is more economical than resolution at the discourse level.

At the same time, while both reflexives are subject to the locality constraint, they seem to differ in the timing of when the locality constraint is applied when appearing in subject position (see more in Sect. 6.1). Upon encountering the reflexive, participants exhibited an immediate locality bias for *ta-ziji*, whereas the ambiguity of *ziji* persisted until the sentence-final position. This finding aligns with broader evidence that different anaphoric forms may display distinct sensitivities to a shared constraint within the same language (e.g., Kaiser & Trueswell, 2008; Patil et al., 2023). That said, we acknowledge a reviewer’s point that the absence of a reliable Distance $\times$ Reflexive interaction at the critical region limits the strength of the evidence for an early form-specific difference. Accordingly, we treat this difference as suggestive rather than conclusive. Nevertheless, the two reflexives diverge at the subsequent

⁶ While sentence-final “wrap-up” effects have sometimes been discounted, there is substantial evidence that these so-called “wrap-up” effects often reflect meaningful integrative processes such as resolution of ambiguities in sentence comprehension (e.g., Just & Carpenter, 1980; Rayner et al., 2000; Hirotsu et al., 2006; Payne & Stine-Morrow, 2014; Stowe et al., 2018). As Stowe et al. (2018: 245) noted, “the embargo on sentence-final positions has likely blocked research on important effects.” At the same time, we agree with an anonymous reviewer that sentence-final effects should be interpreted with caution. Such effects may reflect sentence-level integration, reanalysis, or other wrap-up processes that are distinct from the online binding computations reflected at preceding regions. Therefore, locality effects observed at the sentence-final region may not necessarily provide the same type of evidence for real-time binding preferences as effects observed at the reflexive or spillover regions. Exploring the relationship between wrap-up effects and earlier processing effects is a valuable direction for future work.

spillover region with respect to the locality effect, indicating that some degree of form-specific differentiation emerges during processing.

With respect to the strength of the locality bias, our findings do not provide strong evidence that one reflexive is more constrained by locality than the other. Both reflexives exhibit locality effects in post-reading acceptability judgments, and both show locality biases during online processing, albeit at different stages. Although *ta-ziji* appeared to show an earlier locality effect, we refrain from equating the timing of an effect with its strength. Our conclusions thus differ from previous studies that reported a stronger locality bias for *ziji* (Wang, 2017) or for *ta-ziji* (Lu, 2011; Ma & Wu, ms.), as well as from Ma and Wu (2025) who found no difference in the timing of the locality bias. These discrepancies may be attributed to methodological factors. For example, previous studies often placed reflexives after verbs that could take direct objects (e.g., 'like' or 'suspect'), which could lead readers to initially parse the reflexive as an object (Lu, 2011; Wang, 2017; Ma & Wu, 2025). In addition, Lu (2011) and Wang (2017) tested relatively small samples (32~48 participants), while Wang (2017) and Ma and Wu (2025) put reflexives inside relative clauses, rendering the 'local' antecedent structurally more embedded than in our design.

Finally, we consider an alternative explanation: the differences between *ziji* and *ta-ziji* may be related to their potential use as intensifiers. As noted by Wang (2017: 531) and Dillon et al. (2016: 13), *ta-ziji* in subject position can receive an intensifier interpretation, which could account for the distinct processing patterns observed. A similar possibility may also hold for *ziji*. To evaluate this hypothesis, Experiment 2 introduces a *shi*-cleft construction in which *ziji/ta-ziji* immediately follows the focus particle *shi*, thereby enforcing an intensifier interpretation.

Experiment 2: Processing of Intensifiers

Participants

Participants included one hundred and four native speakers with normal or corrected-to-normal vision from Peking University. Informed consent was obtained from participants who were compensated for participation. None had taken part in Experiment 1.

Design and Materials

The experimental design in Experiment 2 is identical to Experiment 1, except in Experiment 2, the reflexive is placed after the focus particle *shi* (see (7) in Sect. 2.2). All target items were minimally adapted from those in Experiment 1 by adding the *shi* particle before the reflexive. The filler items were identical.

Procedure

The procedure was identical to that of Experiment 1.

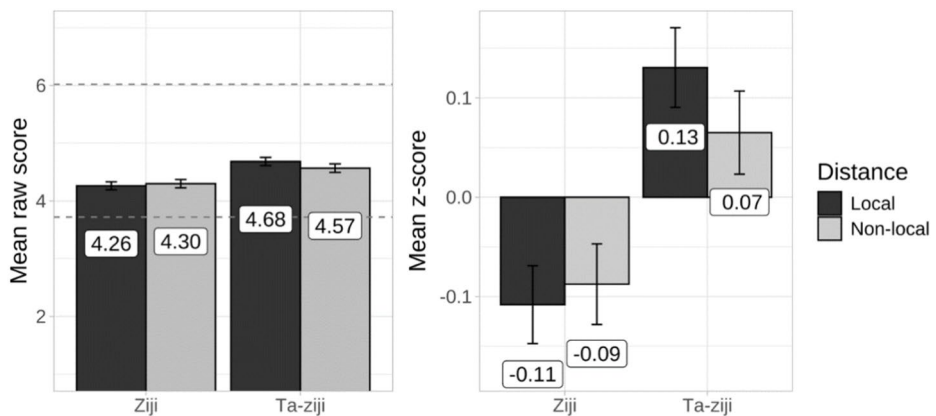


Fig. 3 Mean acceptability ratings for target conditions in Experiment 2. The left panel displays raw scores; the right panel displays z-transformed scores. Error bars indicate standard errors. The two dashed lines on the left panel represent the mean ratings for grammatical and ungrammatical fillers, respectively

Table 3 Summary of statistics for acceptability judgments in Experiment 2 (*: $p < 0.05$)

	β	SE	t	p
Distance	0.06	0.11	0.54	0.59
Reflexive	-0.62	0.15	-4.13	<0.001*
Distance x Reflexive	-0.24	0.15	-1.57	0.12

Model structure: $\text{clmm}(\text{AJ} \sim \text{Distance} * \text{Reflexive} + (1 + \text{Reflexive} | \text{Item}) + (1 + \text{Reflexive} + \text{Distance} | \text{Participant}), \text{data})$

Data analysis

Data were analyzed using the same methods as specified in Experiment 1. Three participants were excluded before data analysis as they rated ungrammatical fillers as being more acceptable than grammatical fillers. Thus, the final analysis included 101 participants.

Results

Acceptability Judgments

Figure 3 displays participants' acceptability judgments for the target conditions. See Table 3 for statistics. The analysis revealed a main effect of Reflexive ($p < 0.001$) as sentences with *ta-ziji* were rated as more acceptable than those with *ziji*. No main effect of Distance or Reflexive x Distance interaction was observed. The absence of a Distance effect, contrasting with the locality effect observed in Experiment 1, suggests that the presence of the focus particle *shi* modulates binding preferences for *ziji* and *ta-ziji*.⁷ This reduced locality bias is

⁷ Because the effect sizes were very small in the two experiments, a combined analysis comparing the Distance effect across the two experiments did not reveal any significant difference in locality biases (see supplementary materials for a summary of statistics). However, as noted below, the binding preferences differed in online processing between the two experiments.

consistent with Kaiser and Runner's (2008) findings in Dutch, where the intensifier *hemzelf* showed a weaker locality bias than the plain anaphor *zichzelf*.

Reading Times

Figure 4 displays participants' mean reading times by region and condition (see Table 4 for statistics). Prior to the critical reflexive region, no effects reached significance. At the reflexive, we found a main effect of Reflexive ($p < 0.001$), with *ta-ziji* eliciting longer reading times. Neither the Distance effect nor the Distance x Reflexive interaction was significant. At the first spillover region (BA), no main effect was observed, but the Distance x Reflexive interaction was significant ($p = 0.01$). Pairwise comparisons showed that *ziji* exhibited a locality bias ($\beta = -0.04$, $SE = 0.02$, $t = -2.17$, $p = 0.03$), whereas *ta-ziji* showed no reliable binding preference ($\beta = 0.03$, $SE = 0.02$, $t = 1.48$, $p = 0.16$), despite a numerical trend toward a non-local bias. No significant effects were observed in subsequent regions.

Given these results, it appears that the processing of intensifiers in Experiment 2 differs sharply from that of plain anaphors in Experiment 1. The difference is especially evident for *ta-ziji*: when interpreted as a plain anaphor, it strongly favors a local antecedent, but when parsed as an intensifier, the local binding preference disappears. The pattern for *ziji* also shifts, with the locality bias emerging earlier under the intensifier reading. To further examine the role of contrastive focus, we conducted a combined analysis by including Experiment as a between-participants factor, which allows us to directly compare processing patterns across the two experiments. Below, we only report statistical effects that reveal differences in reflexive processing between the two experiments (i.e., two/three-way interactions between the factor Experiment and the independent variables). A complete summary of statistics is available in supplementary materials.

The combined analysis for the critical reflexive region revealed a Distance x Experiment interaction ($\beta = -0.05$, $SE = 0.02$, $t = -2.01$, $p < 0.05$), reflecting a stronger locality bias in Experiment 1 (processing of plain anaphors) than in Experiment 2 (processing of intensifiers). No other interactions were observed at this region. At the first spillover region,

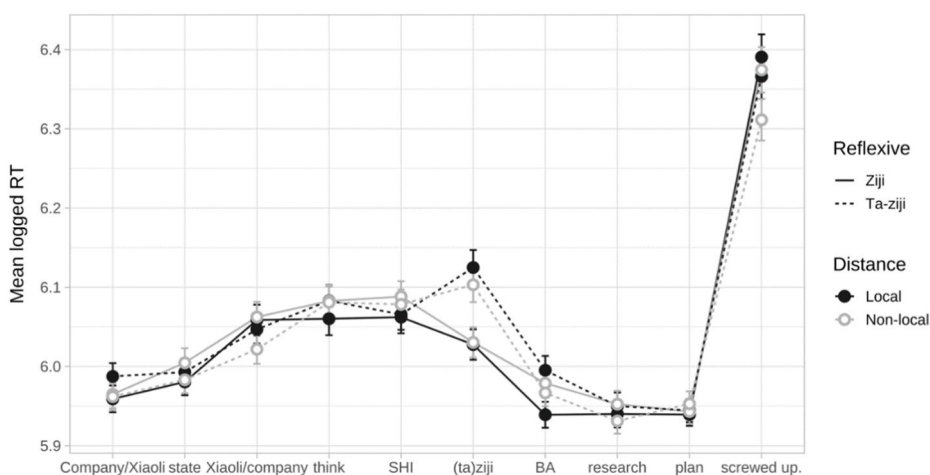


Fig. 4 Mean log-transformed RTs by region and condition in Experiment 2. Error bars represent standard errors

Table 4 Summary of statistics for log-transformed RTs in Experiment 2 (*: < 0.05)

Region	Distance				Reflexive				Distance x Reflexive			
	β	SE	t	p	β	SE	t	p	β	SE	t	p
'Xiaoli/company'	0.01	0.01	0.88	0.38	-0.01	0.01	-0.92	0.36	-0.04	0.02	-1.40	0.16
'state'	-0.009	0.01	-0.74	0.46	0.004	0.01	0.38	0.71	-0.04	0.02	-1.50	0.13
'company/Xiaoli'	0.01	0.01	0.75	0.45	0.02	0.01	1.77	0.08	-0.03	0.03	-1.01	0.31
'think'	-0.008	0.02	-0.53	0.59	-0.01	0.02	-0.57	0.57	-0.02	0.03	-0.65	0.51
<i>shi</i>	-0.02	0.02	-1.24	0.22	0.004	0.02	0.23	0.82	-0.01	0.02	-0.31	0.76
<i>(ta-)ziji</i>	0.008	0.02	0.51	0.61	-0.09	0.02	-5.34	<0.001*	-0.02	0.03	-0.51	0.61
BA	-0.007	0.01	-0.53	0.59	-0.02	0.01	-1.52	0.13	-0.07	0.03	-2.54	0.01*
'research'	0.002	0.01	0.17	0.87	0.006	0.01	0.49	0.62	-0.03	0.03	-1.29	0.20
'plan'	-0.005	0.01	-0.48	0.63	-0.008	0.01	-0.69	0.49	0.003	0.02	0.15	0.88
'screw up'	0.04	0.02	1.80	0.07	0.05	0.02	1.93	0.06	-0.04	0.04	-0.86	0.39

Model structure:

Region 1: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)

Region 2: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)

Region 3: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)

Region 4: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 + Reflexive | Participant), data)

Region 5: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)

Region 6: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)

Region 7: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 | Participant), data)

Region 8: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 + Distance | Participant), data)

Region 9: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 + Reflexive | Participant), data)

Region 10: lmer(logRT ~ Distance * Reflexive + (1 | Item) + (1 + Reflexive | Participant), data)

there was a three-way interaction among Distance, Reflexive, and Experiment ($\beta=0.14$, $SE=0.04$, $t=3.59$, $p<0.001$). This effect arose because the Distance x Experiment interaction differed across reflexive types: the locality bias difference between experiments was only marginal for *ziji* ($\beta=0.05$, $SE=0.03$, $t=1.96$, $p<0.10$) but significant for *ta-ziji* ($\beta=-0.08$, $SE=0.03$, $t=-3.03$, $p=0.002$), with the locality bias for *ta-ziji* present only in Experiment 1, but absent in Experiment 2. No other three-way interactions were detected at the following regions. Finally, at the final region, a Distance x Experiment interaction was observed ($\beta=-0.08$, $SE=0.03$, $t=-2.57$, $p=0.01$), echoing the stronger locality bias in Experiment 1 relative to Experiment 2 at the critical region.

In summary, the combined analysis suggests that the processing of *ziji* does not differ substantially between its anaphoric and intensifier uses, although the locality effect appears to emerge earlier when interpreted as an intensifier relative to its interpretation as a plain anaphor. In contrast, the processing pattern of *ta-ziji* showed a clear shift: when interpreted as a plain anaphor, *ta-ziji* exhibited a strong bias for local binding, but under the intensifier interpretation this locality bias disappeared and even reversed numerically, which highlights the role of contrastive focus in real-time reflexive resolution.

Discussion

Experiment 2 investigates the interpretation and processing of *ziji* and *ta-ziji* in the *shi*-cleft construction, where the reflexives are obligatorily interpreted as intensifiers. The goal is to determine whether and when contrastive focus modulates the processing of Chinese reflexives. This experiment was motivated by the ongoing debate about the accessibility of non-syntactic cues in real-time processing. The syntax-first account posits that non-syntactic constraints are applied only at later processing stages, such that initial reflexive resolution is primarily guided by syntactic considerations, and non-syntactic effects are expected to emerge during later processing stages. Contrastively, a multiple-constraints account predicts that non-syntactic factors, including the contrastive focus examined in the present study, can influence processing since the earliest stages.

The offline acceptability judgments revealed that participants did not systematically prefer either local or non-local binding in Experiment 2 due to the absence of a main effect of Distance. This contrasts with Experiment 1, where local binding was consistently preferred in non-contrastive contexts, suggesting that binding preferences for reflexive intensifiers differ from those for reflexive anaphors. Interestingly, the attenuated locality bias observed in Experiment 2 broadly aligns with prior findings in Dutch, where intensifiers exhibited weaker locality effects compared to plain anaphors (Kaiser & Runner, 2008). This attenuation in locality effects may reflect the fact that either the local or non-local referent can serve as the contrasted entity, thus lowering the likelihood that the local referent is selected as the antecedent.

Reading times from the self-paced reading task, however, revealed a more nuanced picture. We found that the processing of *ta-ziji* was immediately and reliably affected by the introduction of contrastive focus. Unlike in Experiment 1 where a robust locality bias was observed for both reflexives, in Experiment 2 no binding preference was observed for *ta-ziji*. This contrast in processing patterns was confirmed in the combined analysis comparing the two experiments, pointing to early divergence during processing at the onset of *ta-ziji*

as well as at subsequent spillover regions. In contrast, *ziji* consistently showed a locality bias in structural contexts that either bias a plain anaphor reading or an intensifier reading.

Taken together, the early emergence of the contrastive focus effect aligns with the multiple-constraints account, showing that contrastive focus can guide reflexive resolution since the initial stages of processing. Moreover, these results indicate a form-specific influence of contrastive focus on reflexive resolution. Contrastive focus exerts a stronger effect on *ta-ziji* than on *ziji*, especially in online processing. In Sect. 6, we further consider how this form-specific difference in real-time processing bears on different findings on the locality bias between the two reflexives in previous studies (e.g., Dillon et al., 2016; Wang, 2017).

General Discussion

The present study investigated the processing of the Chinese reflexives *ziji* and *ta-ziji* in structural contexts that either bias plain anaphor readings or enforce intensifier readings, using self-paced reading and acceptability judgment tasks. Our first aim was to examine the locality bias for *ziji* and *ta-ziji* in subject position, which has generated considerable debate (e.g., Lu, 2011; Dillon et al., 2016; Wang, 2017; Ma & Wu, 2025). Our second aim was to probe the processing of reflexive intensifiers and, in doing so, to examine the role of contrastive focus in real-time reflexive resolution. To address these questions, we employed an animacy mismatch paradigm to create local or non-local binding contexts for the two reflexives.

Results from end-of-sentence acceptability judgments showed that in Experiment 1, where the structurally closer subject occupies the local position (e.g., Huang et al., 2001), native Chinese speakers preferred local over non-local binding, which suggests that Chinese speakers generally favored interpreting the reflexives as plain anaphors. In contrast, Experiment 2 revealed no reliable locality effects when the contrastive focus particle *shi* preceded the reflexive, thereby enforcing an intensifier interpretation. Thus, the offline results established that the binding preferences of plain anaphors differ from those of intensifiers.

Reading times from the self-paced reading task revealed a more complex picture in two aspects. First, in Experiment 1, while locality biases were observed, which is consistent with offline judgments, the timing of the locality biases differed between the two reflexives: the locality effect for *ziji* appeared delayed relative to that for *ta-ziji*. Second, in Experiment 2, where the reflexives were obligatorily interpreted as intensifiers, *ta-ziji* showed no locality bias, whereas *ziji* exhibited an earlier locality effect relative to Experiment 1. This asymmetry regarding the attenuation of the locality effects for *ziji* and *ta-ziji* across the two experiments suggests that contrastive focus plays a more prominent role in the processing of *ta-ziji* than in the processing of *ziji*. Below, we discuss each of these two findings in turn.

Different Processing Patterns of *Ziji* and *Ta-ziji* as Plain Anaphors

Reading times from Experiment 1 showed that Chinese speakers immediately formed local dependencies upon encountering *ta-ziji* but seemed to delay resolution with *ziji*. This finding differs from previous studies which either reported asymmetrical binding preferences (e.g., Lu, 2011; Wang, 2017) or did not find any form-specific difference in the timing of the locality biases (Ma & Wu, 2025). As noted in Sect. 2.1, this could be attributed to dif-

ferences in experimental design. For example, in Wang (2017) and Ma and Wu (2025), the reflexive was embedded inside a relative clause and could be initially parsed as a direct object. Another possibility is that verb semantics could modulate the interpretation patterns, especially for *ziji* as shown by previous research (e.g., Lu, 2011; Liu, 2020; Lyu & Kaiser, 2021, 2023).

We argue that verb information is particularly important for the processing of *ziji* in subject position. While a large body of theoretical and experimental research suggests that non-local binding can be closely tied to perspective-taking, i.e., the speaker or reader of the sentence needs to adopt the perspective of the sentence-internal perspectival center (e.g., Huang et al., 2001; Pan, 2001; Charnavel et al., 2017; Liu, 2020; Lyu & Kaiser, 2021, 2024), other non-perspectival factors are equally important for the interpretation of *ziji* (e.g., Yu, 1992; Pollard & Xue, 1998; Hu, 2019; Sun & Li, 2023). These include world knowledge and verb semantics, which can override perspective-taking (e.g., Yu, 1992; Pollard & Xue, 1998; Hu, 2019; Sun & Li, 2023).

Yu (1992) demonstrated this interaction with examples such as ‘the president₁ asked me₂ to sit beside self_{1/*2}’ and ‘Lisi₁ worried that I₂ might surpass self_{1/*2}’. In both sentences, the local antecedent is a first-person pronoun which typically enforces local binding, a phenomenon called “the blocking effect.” The blocking effect arises because speakers or readers prioritize their own perspectives rather than others’ (e.g., Kuno, 1987; Huang et al., 2001). However, in these sentences, the verb information overrides the blocking effect because non-local binding is pragmatically more plausible according to world knowledge. Thus, when *ziji* occurs in subject position, the parser may delay commitment to binding until the verb is encountered, since the predicate is crucial for resolving ambiguity. By contrast, *ziji* in object or possessor position often shows no significant delay in displaying a locality effect (e.g., Chen et al., 2012; Dillon et al., 2014, 2016; Jäger et al., 2015; Lyu & Kaiser, 2023) presumably because the predicate precedes the reflexive. As for *ta-ziji*, it appears less sensitive to verb semantics relative to *ziji* (Lu, 2011: 35) and thus verb semantics may play a less prominent role, which could account for the early locality effect observed for *ta-ziji* in Experiment 1. Future research is required to shed light on whether the position of the verb affects the time-course of the locality effect.

Different Processing Patterns of *Ziji* and *Ta-ziji* as Intensifiers

A key finding of Experiment 2 is that in the *shi*-cleft construction, where both reflexives are obligatorily parsed as intensifiers, the strong locality bias observed for *ta-ziji* in Experiment 1 disappeared. In fact, we even observed a numerically trending anti-locality bias at two of the spillover regions. For *ziji*, the locality bias emerged earlier in Experiment 2 than in Experiment 1 where it was preferentially interpreted as a plain anaphor. However, since the combined analysis showed that the shift in processing patterns was statistically reliable only for *ta-ziji*, we refrain from making strong claims about *ziji*.

The shift in processing patterns for *ta-ziji* at the critical region provides clear evidence that contrastive focus can immediately modulate the resolution of reflexives. This finding is consistent with Baker’s (1995) theoretical account of intensifier reflexives where the prominence of candidate antecedents plays a crucial role. Specifically, as the matrix subject is arguably more prominent in discourse by virtue of being a discourse topic (see e.g., Li & Thompson, 1981), it can compete with the syntactically prominent local subject. Conse-

quently, the locality bias was substantially attenuated in Experiment 2. Crucially, the fact that this attenuation occurs immediately at the onset of the reflexive aligns with the predictions of the multiple-constraints account, which holds that non-syntactic cues – such as the contrastive focus particle *shi* in this case – can be recruited at the earliest moments of reference resolution. Similar support for this account has been reported in prior psycholinguistic research on reflexives, particularly in Chinese (e.g., Chen et al., 2012; Chang et al., 2020; Lyu & Kaiser, 2021, 2023), where semantic (e.g., gender) and discourse cues (e.g., discourse topicality, perspective-taking) have been shown to guide online reflexive resolution.

However, one open question concerns why *ziji* and *ta-ziji* are differentially sensitive to contrastive focus. It appears that contrastive focus has a larger impact on the processing of *ta-ziji* compared to *ziji*. Although this question is not crucial for the purposes of this study as our goal is primarily to explore the accessibility of contrastive focus, we tentatively speculate that it may be tied to the different discourse properties of the two reflexives. In our experiment, the matrix subject always is the subject of speech verbs, while the local subject is always the subject of belief verbs. Thus, the differences between *ziji* and *ta-ziji* as intensifiers could reflect their varying sensitivities to discourse roles. In particular, *ta-ziji* as an intensifier may be more likely than *ziji* to refer to subjects of speech verbs (see Lyu & Kaiser, 2023, 2024, for evidence that discourse factors could affect the two reflexives differently). Future research should explore this possibility further.

Finally, we would like to comment on the different findings in previous research regarding the relative strength of the locality bias for *ziji* and *ta-ziji* in light of our results on intensifiers. As noted in Sect. 2.1., Dillon et al. (2016) and Wang (2017) reported stronger locality bias for *ziji* than for *ta-ziji*. They suggested that one of the reasons for the weaker locality bias for *ta-ziji* in their study could be due to its potential interpretation as an intensifier when it occurs in subject position. In other words, Chinese speakers may be more likely to treat *ta-ziji* as an intensifier, thereby reducing its locality bias, since long-distance binding of intensifier *ta-ziji* across clauses and sentences is allowed (e.g., Pan, 1998). Our findings provide empirical support for this hypothesis: when *ta-ziji* is interpreted as an intensifier, its binding preferences change substantially. The methodological implication, therefore, is that comparative studies of plain anaphors should also consider their potential interpretations as non-syntactic anaphors. We view our research as one step in this direction.

Conclusion

The present study examined the processing of *ziji* and *ta-ziji* in structural contexts that bias either plain anaphor or intensifier readings, combining evidence from acceptability judgments and self-paced reading times. Results from both experimental tasks indicate that the binding preferences of the two reflexives are modulated by contrastive focus. This effect is particularly prominent for *ta-ziji*, which exhibits a strong locality bias when interpreted as a plain anaphor but shows no clear bias when construed as an intensifier. Importantly, this shift in binding preference emerges immediately at the onset of *ta-ziji*, which aligns more with a multiple-constraints approach to online reflexive resolution in Chinese. Moreover, contrastive focus affects the processing patterns of the two reflexives to different degrees, suggesting that prior debates on differences in locality biases between *ziji* and *ta-ziji* should take into account the likelihood of intensifier readings.

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Data Availability The data, scripts for statistical analyses, and supplementary materials are available at <https://doi.org/10.17605/OSF.IO/GK3YX>.

Declarations

Conflict of Interest The author has no conflicts of interest to declare.

Ethical Approval All procedures performed in this study were in accordance with the 1964 Helsinki declaration and its later amendments. Informed consent was obtained from participants included in the study.

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